

# CUONG PHAM (Phạm Anh Cường)

◇ Address: Abu Dhabi, United Arab Emirates  
◇ Email: [phamanhcuongvna2511@gmail.com](mailto:phamanhcuongvna2511@gmail.com)  
◇ Portfolio: [pacman-ctm.github.io](https://pacman-ctm.github.io)

◇ Phone: (+971) 58 520 6140  
◇ GitHub: [github.com/pacman-ctm](https://github.com/pacman-ctm)  
◇ LinkedIn: [linkedin.com/in/anhcuong-pham](https://linkedin.com/in/anhcuong-pham)

## PROFILE

Master's student in Machine Learning with research interest in **Optimization** in general and **Efficient Deep Learning** in particular. In detail, I am primarily working on **Efficient Training/Fine-Tuning** at the intersection of **Continual Learning**, **Federated Learning**, **Large models**, and partially on **Optimization for Machine Learning**.

I am actively looking for an RA/RE position in Continual Learning/Optimization/Efficient Deep Learning this year, and preparing for 2027 PhD application.

## EDUCATION

**Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)**, UAE 2024 - 2026  
*Master of Science in Machine Learning* GPA: 3.72/4.0 (~ 94%) [Transcript]

- Selected courses: Mathematics for AI, Advanced Machine Learning, Probabilistic & Statistical Inference, Optimization.
- Research topics: Continual Learning, Efficient Deep Learning, Optimization for Machine Learning.
- Thesis: "Learning in the Null Space: Small Singular Values for Continual Learning".
- Supervisors: [Prof. Samuel Horváth](#) and [Prof. Praneeth Vepakomma](#) (informally).

**VNU University of Engineering and Technology (VNU-UET)**, Vietnam 2019 - 2023  
*Bachelor in Information Technology (Honors Program)* GPA: 3.71/4.0 (8.78/10) [Transcript]

- Graduated with High Distinction, top 5% of CS Department.
- Thesis: "Building a Vietnamese image captioning dataset and developing a visual-text transformation model for generating image captions" (9.5/10).
- Supervisors: [Prof. Quang-Thuy Ha](#), [Dr. Van-Quang Nguyen](#), [MS. Thi-Hong Vuong](#).

## PUBLICATIONS

(P = preprint, W = workshop, C = conference, J = journal, R = project report, O = ongoing paper)

[O.6] **Cuong Anh Pham**, Praneeth Vepakomma, Samuel Horváth. "TBD" (ongoing draft).

[C.5] **Cuong Anh Pham**, Praneeth Vepakomma, Samuel Horváth. "Learning in the Null Space: Small Singular Values for Continual Learning" (CPAL '26, **Oral Presentation**).

[P.4] Farshed Abdukhakimov, **Cuong Anh Pham**, Samuel Horváth, Martin Takáč, Slavomir Hanzely. "Polyak Stepsize: Estimating Optimal Functional Values Without Parameters or Prior Knowledge" (preprint 2508.17288, under-review draft).

[P.3] Quang P.M. Pham, Khoi T. N. Nguyen, Nhi H. Doan, **Cuong A. Pham**, Qinbo Sun, Weimin Qi, Kentaro Inui, Dezhen Song. "SmallPlan: Leverage Small Language Models for Sequential Path Planning with Simulation-Powered, LLM-Guided Distillation" (preprint 2505.00831).

[P.2] **Anh-Cuong Pham**, Van-Quang Nguyen, Thi-Hong Vuong, Quang-Thuy Ha. "KTVIC: A Vietnamese Image Captioning Dataset on the Life Domain" (preprint 2401.08100, short version of Bachelor's thesis).

[C.1] Quynh-Trang Pham Thi, **Anh-Cuong Pham**, Ngoc-Huyen Ngo and Duc-Trong Le. "Memory-Based Method using Prototype Augmentation for Continual Relation Extraction" (IEEE RIVF '22).

## EXPERIENCE (Research)

**MBZUAI**, Abu Dhabi, UAE 10/2024 - Present  
*Graduate Research Assistant*

- **(Thesis) Project:** Efficient Training/Fine-Tuning for Continual Learning (ongoing - Coauthor paper [C.5])  
*Supervisors:* [Prof. Samuel Horváth](#) and [Prof. Praneeth Vepakomma](#)  
Developing an algorithm that can preserve the performance of previous tasks in Continual Learning using orthogonality. Ongoing extension including Efficient Continual Fine-Tuning (ongoing paper [O.6]) and Federated Continual Learning. Skills: Python, Pytorch | Continual Learning, Federated Learning, Parameter-Efficient Fine-Tuning, SVD, LoRA.

- **Project:** Parameter-free SGD Optimization with Polyak stepsize (Coauthored paper [P.4])  
*Supervisors:* Prof. Samuel Horváth and Prof. Martin Takáč  
(Supported seniors to) Develop an adaptive Polyak-based stepsize for SGD Optimization that does not require approximation for optimal value, with the target to reduce the need for hyperparameter tuning during the training process.  
Skills: Python, Scikit-learn, Pytorch | Optimization.
- **Project:** Small Language Models application for edge-device Robot Learning (extended from the coursework project at MBZUAI, Coauthored paper [P.3])  
(Supported seniors to) Utilizing SLMs with the guidance from LLMs to enable efficient and adaptive real-time inference on edge devices for autonomous robotics path-planning tasks.  
Skills: Python | LLMs.

**DSKT Laboratory at VNU-UET**, Hanoi, Vietnam

09/2021 - 09/2023

*Student Research Intern*

- **(Thesis) Project:** Image Captioning (in Vietnamese context) (Coauthored paper [P.2])  
*Mentors:* Dr. Van-Quang Nguyen and Prof. Quang-Thuy Ha  
Created a new image-caption dataset in Vietnamese and developed a tiny Image Captioning model based on the Transformer framework. The dataset and tiny model were evaluated with variants of Deep Learning-based models.  
Skills: Python, Pytorch, OpenCV | Computer Vision, NLP.
- **Project:** Applying Prototype Augmentation for Continual Relation Extraction (Coauthored paper [C.1])  
*Mentors:* MS. Quynh-Trang Thi Pham and Dr. Duc-Trong Le  
(Supported seniors to) Apply a prototype augmentation mechanism to consistent representation learning to reduce saved data for the next phase of continual learning, but still alleviate the forgetting problem.  
Skills: Python | Continual Learning, NLP.

## EXPERIENCE (Industry)

**Viettel Networks**, Hanoi, Vietnam

10/2023 - 07/2024

*Data Scientist*

- Researched and applied **Computer Vision** (image processing, segmentation, classification) for tasks with telecommunication images to automate processes instead of using manual work for network maintenance.
- Solved **Big Data** problems: using Machine Learning with GPUs with large-scale telecommunication data (100M to 10B records), optimizing and automating processes by predicting users' usage and assisting in the allocation of network bandwidth to improve network users' experience.

**GHTK Joint Stock Company**, Hanoi, Vietnam

*R&D Engineer*

08/2022 - 08/2023

- Researched and developed (team of 2 persons) a mail spam filtering based on **Deep Learning** models, and **NLP** frameworks (BERT, DeBERTa, PhoBERT), achieving the f1-score of 90% after training with 50k emails, it was then deployed to the company's internal email system.
- Participated in building an internal chatbot using rule-based techniques.

*R&D Engineer Intern*

05/2022 - 08/2022

- MasterDev Season 4 Internship Program in Research and Development domain. Training about Probability & Statistics, Natural Language Processing, and Big Data (Hadoop and Spark).
- Final Project about building a text classification for Vietnamese documents using doc2vec.

## INVITED TALKS

**"Learning in the Null Space: Small Singular Values for Continual Learning"**

- Oral Presentation at The Third Conference on Parsimony and Learning (Tübingen, Germany) 2026
- RelationalML Group at CISPA, hosted by Prof. Rebekka Burkholz (online) 2026

## HONORS & AWARDS

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|--------------------------------------------------------------------------|------|
| Full scholarship for 2 years of the Master of Science program at MBZUAI  | 2024 |
| Third Prize (5th place) in Linear Algebra, VNU-UET Mathematical Olympiad | 2023 |
| Third Prize (6th place) in Linear Algebra, VNU-UET Mathematical Olympiad | 2022 |
| Yamada Scholarship (6 students/year at VNU-UET)                          | 2022 |
| Third Prize (5th place) in Linear Algebra, VNU-UET Mathematical Olympiad | 2021 |

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|----------------------------------------------------------------------------------------------|------|
| Excellent Scholarship for top highest GPA from VNU-UET (rank 5th)                            | 2021 |
| Attended Vietnam Mathematical Olympiad for high school students (among top 250/600 students) | 2019 |

## CONFERENCES, SCHOOLS & SEMINARS ATTENDED

|                                                                                         |              |
|-----------------------------------------------------------------------------------------|--------------|
| Efficient Machine Learning Seminar at MBZUAI (Abu Dhabi, UAE)                           | 2025-present |
| (CPAL '26) The Third Conference on Parsimony and Learning (Tübingen, Germany)           | 2026         |
| (MLWS '26) MBZUAI Machine Learning Winter School (Abu Dhabi, UAE)                       | 2026         |
| (ICOMP '25) The International Conference on Computational Optimization (Abu Dhabi, UAE) | 2025         |
| (ASCOMP '25) The AI Autumn School of Computational Optimization (Abu Dhabi, UAE)        | 2025         |

## SKILLS

**Programming:** Python, Java, C, C++, HTML/CSS, JavaScript, MySQL.  
**Technologies:** PyTorch, Scikit-learn, Linux, Docker, Hadoop, Spark, Laravel.  
**Additional:** Microsoft Office, L<sup>A</sup>T<sub>E</sub>X, Git.  
**Language:** English (with TOEIC LR 855, IELTS 7.0), Vietnamese (native).

## REFERENCES

|                                                                                            |                               |
|--------------------------------------------------------------------------------------------|-------------------------------|
| <b>Dr. Samuel Horvath</b> , Assistant Professor at MBZUAI (UAE):                           | (samuel.horvath@mbzuai.ac.ae) |
| <b>Dr. Praneeth Vepakomma</b> , Assistant Professor at MBZUAI & Visiting Professor at MIT: | (vepakom@mit.edu)             |
| <b>Dr. Ha Quang Thuy</b> , Associate Professor & Former Vice Rector at VNU-UET (Vietnam):  | (thuyhq@vnu.edu.vn)           |
| <b>Dr. Nguyen Van Quang</b> , Researcher at RIKEN AIP (Japan):                             | (quang.nguyen.jz@riken.jp)    |